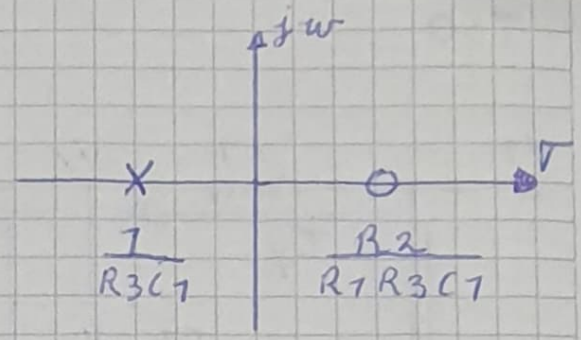
Diagrama de polos y ceros

$$H(s) = \frac{s - \frac{R2}{R1R3C1}}{s + \frac{1}{R3C1}} \longrightarrow$$



Cálculo de módulo y fase

$$H(w) = H(s)\Big|_{s=jw} = \frac{jw - \frac{R2}{R1R3C1}}{jw + \frac{1}{R3C1}}$$

$$|H(w)| = \left|\frac{jw - \frac{R2}{R1R3C1}}{jw + \frac{1}{R3C1}}\right| = \frac{\sqrt{w^2 + \left(\frac{R2}{R1R3C1}\right)^2}}{\sqrt{w^2 + \frac{1}{(R3C1)^2}}}$$

$$\boxed{|H(w)| = \sqrt{\frac{w^2 - \left(\frac{R2}{R1R3C1}\right)^2}{w^2 + \frac{1}{(R3C1)^2}}}}$$

$$\theta(w) = \measuredangle H(w) \longrightarrow H(w) = \frac{|H_A(w)|}{|H_B(w)|} \cdot \frac{e^{j\theta_A(w)}}{e^{j\theta_B(w)}}$$

$$|H(w)| \qquad e^{j(\theta_A(w) - \theta_B(w))}$$

$$\rightarrow \theta(w) = \operatorname{arc\,tg}\left(\frac{w}{\frac{R2}{R1R3C1}}\right) - \operatorname{arc\,tg}\left(\frac{w}{\frac{1}{R3C1}}\right)$$

$$\rightarrow \boxed{\theta(w) = \operatorname{arc\,tg}\left(\frac{R1R3C1 \cdot w}{R2}\right) - \operatorname{arc\,tg}(R3C1 \cdot w)}$$